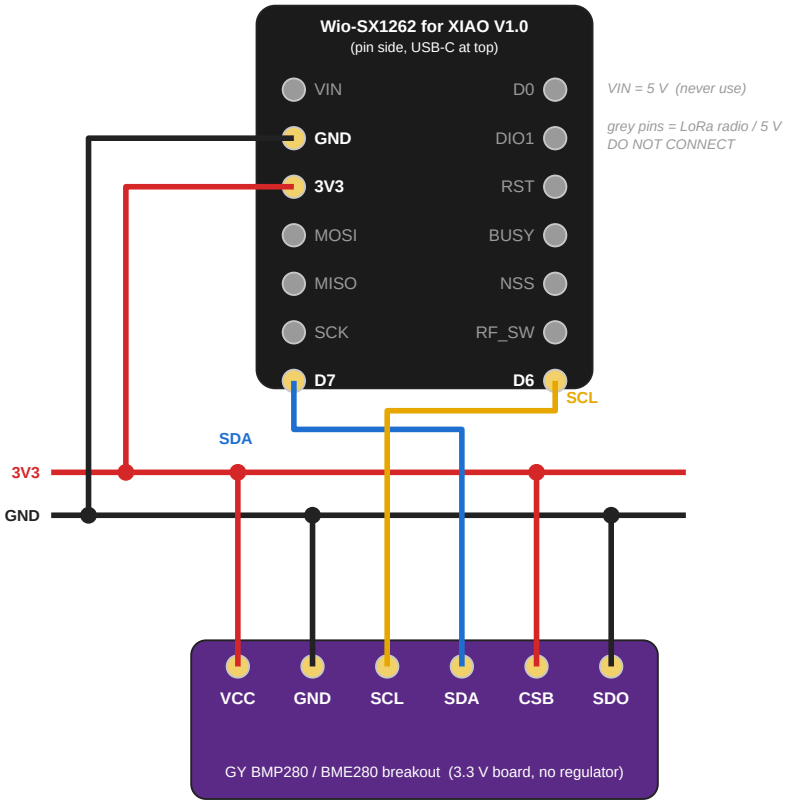
BMP280 → XIAO nRF52840 Wiring (MeshCore)

Pin mapping verified against MeshCore `variants/xiao_nrf52/platformio.ini`: **PIN_WIRE_SCL = D6, PIN_WIRE_SDA = D7**. Prepared 2026-10-08.



Dot = connected. Lines crossing without a dot are NOT connected. Sensor pins drawn left->right in the order printed top->bottom on the board.

Pin-to-pin connections

#	BMP280 pin	XIAO / Wio pin	Wire	Why
1	**VCC**	**3V3 (left row, 3rd)**	**red**	3.3 V power. Not VIN (5 V).
2	**GND**	**GND (left row, 2nd)**	**black**	Ground
3	**SCL**	**D6 (right row, bottom)**	**yellow**	I2C clock
4	**SDA**	**D7 (left row, bottom)**	**blue**	I2C data
5	**CSB**	**3V3**	**red**	High = I2C mode (low = SPI, not detected)
6	**SDO**	**GND**	**black**	Sets address 0x76 (MeshCore default)

Notes

• **SDA and SCL are swapped compared with the Meshtastic layout.** On Meshtastic's `_i2c` build it was SDA=D6, SCL=D7. On MeshCore it is **SCL=D6, SDA=D7**.

• **3.3 V only.** The purple 6-pin board passes VCC straight to the chip, and the nRF52840 pins are not 5 V tolerant.

• **Leave every grey pin unconnected** (VIN, MOSI, MISO, SCK, D0, DIO1, RST, BUSY, NSS, RF_SW). They belong to the LoRa radio or 5 V.

• **The pull-ups are already on the breakout**, so you don't need to add resistors.

• **BMP280 or BME280?** The firmware is built with both `ENV_INCLUDE_BMP280=1` and `ENV_INCLUDE_BME280=1`. Both use 0x76 and the chip ID decides at boot. A BME280 adds humidity.

• **Before powering up:** check with a meter that 3V3 and GND are not shorted, and that nothing is connected to VIN.